Deformation space and cotangent complex $\mathbb{L}_{X/k}$

Motivation cotangent sheaf Ω_X is not sufficient for deformation theory when X is singular.

Reason When X is singular, Ω_X is not "free", and we need a "free resolution".

When $X = \operatorname{Spec} k[x_1, \dots, x_n]/(f_1, \dots, f_r) = \operatorname{Spec} A$

$$\Omega_X = \bigoplus_A A\,dx_1 \oplus \cdots \oplus \bigoplus_A A\,dx_n / \langle df_1, \dots, df_r \rangle$$

A-mod

but there are some relations among f_i's which Ω_X don't capture (syzygy broken).

Setting $\operatorname{char} k = 0$ $\quad A = k[x_1, \dots, x_n]/(f_1, \dots, f_r)$

Goal: ~~Proof~~ Prove that

$$T_{\operatorname{Spec} A}\,\mathrm{Def} = \mathrm{Ext}^1_A(\mathbb{L}_{A/k}, A)$$

$\| $ define as

$$\mathrm{Def}_A(k[\varepsilon]/(\varepsilon^2))$$

$R = k[\varepsilon]/(\varepsilon^2)$

$\|$

$$\left\{ \mathcal{X}/\operatorname{Spec} R \;\middle|\; \begin{array}{l} \mathcal{X}/\operatorname{Spec} R \text{ flat}, \\ X \text{ is the special fiber} \end{array} \right\}$$

Left side

$$\begin{array}{ccc} X & \hookrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ \operatorname{Spec} k & \hookrightarrow & \operatorname{Spec} R \end{array} \qquad \begin{array}{ccc} k[x_i]/(f_j) & \leftarrow & R[x_i]/(f_j+\varepsilon g_j) \\ \uparrow & & \uparrow \\ k & \leftarrow & R \end{array}$$

[Vakil 17, 24.4.D] $M \in \mathrm{Mod}(k[\varepsilon]/(\varepsilon^2))$

Rmk M is flat $\Leftrightarrow$ $\varepsilon : M/\varepsilon M \longrightarrow \varepsilon M$ is an iso

In our example, $M = R[x_1, \dots, x_n]/(f_1+\varepsilon g_1, \dots, f_r+\varepsilon g_r)$ is called the first order family

only with $\{$ flatness

we can call it deformation

Apply the ~~first order~~ rmk to the $M = R[x_i]/(f_j+\varepsilon g_j)$

M is flat

$\forall\ h \in R[x_i]$, if $\varepsilon h = 0$ in M $\Rightarrow$ $h \in \varepsilon M$

$\forall\ h \in R[x_i]$, if $\exists h_i \in R[x_i]$ $\varepsilon h = \sum_i h_i (f_i + \varepsilon g_i)$ $\Rightarrow$ $\exists\ \tilde{h} \in R[x_i]$, $h = \varepsilon \tilde{h} + (f_i)$ in $R[x_i]$

Apply the rmk to $M = R[x_i]/(f_j + \varepsilon g_j)$,

M is flat

$\Leftrightarrow$

$\varepsilon : M/\varepsilon M \longrightarrow \varepsilon M$ is inj

$\Leftrightarrow$

$\forall\, m \in M$, if $\varepsilon m = 0$, then $m \in \varepsilon M$

$\Leftrightarrow$

$\forall\, h + \varepsilon h' \in R[x_i]$, if $\varepsilon(h+\varepsilon h') = 0$ in M, then $h + \varepsilon h' \in \varepsilon M$

$\Leftrightarrow$

$\forall\, h \in k[x_i]$, if $\varepsilon h = 0$ in M, then $h \in \varepsilon M$

$\Leftrightarrow$

$\forall\, h \in k[x_i]$, if $\exists\, h_j + \varepsilon h_j' \in R[x_i]$, $\varepsilon h = \sum_j (h_j + \varepsilon h_j')(f_j + \varepsilon g_j)$, in $R[x_i]$,

then $h \in \varepsilon M$

$\Leftrightarrow$

$\forall\, h \in k[x_i]$, if $\exists\, h_j, h_j' \in k[x_i]$

$$\begin{cases} \sum_j h_j f_j = 0 \\ h = \sum h_j' f_j + h_j g_j \end{cases}$$

, then $h \in \varepsilon M$

$\Leftrightarrow$

~~$\forall\, h \in k[x_i]$~~, if $\exists\, h_j, h_j' \in k[x_i]$, $\sum_j h_j f_j = 0$,

then ~~$\sum_j (h_j' f_j + h_j g_j) \in \varepsilon M$~~

$\sum_i h_i g_j \in (f_i)$

Let $h \in k[x_i]$

$h \in \varepsilon M$.

$\Leftrightarrow \exists\ m \in M,\quad h - \varepsilon m = 0$ in M

$\Leftrightarrow \exists\ t + \varepsilon t' \in R[x_i],\quad h - \varepsilon(t+\varepsilon t') \in (f_j + \varepsilon g_j)$

$\Leftrightarrow \exists\ t \in k[x_i],\quad h - \varepsilon t \in (f_j + \varepsilon g_j)$ in $k[x_i]$

$\Leftrightarrow h \overset{\epsilon}{\in} (f_i)$.

$\Updownarrow$ satisfying

$\forall\ h_j \in k[x_i]$ s.t. $\sum_j h_j f_j = 0$, we have $\sum h_j g_j \in (f_j)$

E.g. Take $A = k[x,y]/(xy)$.

$M = R[x,y]/(xy+\varepsilon_0 g_1, \varepsilon g_2)$, then

M is flat over A

$\Leftrightarrow \forall\ h_1, h_2 \in k[x,y]$, $h_1 \cdot xy + h_2 \cdot 0 = 0$

$\Rightarrow h_1 g_1 + h_2 g_2 \in (xy)$

$\Leftrightarrow g_2 \in (xy)$

It is roughly that $g_2 = 0$, as

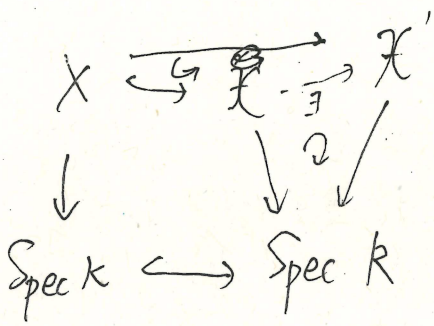
$R[x,y]/(xy+\varepsilon g_1, \varepsilon xy) \cong R[x,y]/(xy+\varepsilon g_1)$

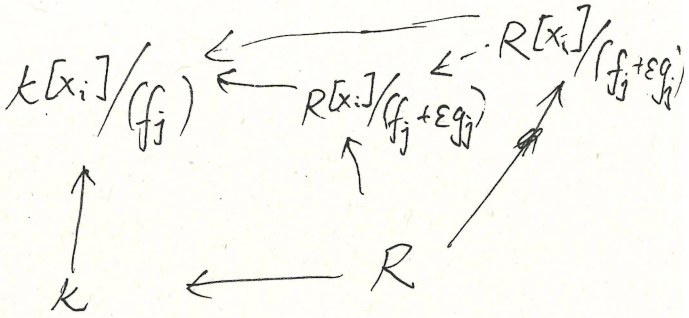
We have provided a lot of deformations of $\operatorname{Spec} A$, but some deformations are essentially the same.

Q: Does

$$R[x,y]/(xy+\varepsilon) \quad \& \quad R[x,y]/(xy+\varepsilon(1+x))$$

essentially provides the same deformation?





If so, the comm diagram requires that

$$\varepsilon \longmapsto \varepsilon$$
$$x_i \longmapsto x_i + \varepsilon h_i$$

E.g. $A = k[x,y]/(xy)$,

$$\varepsilon \longmapsto \varepsilon$$
$$x \longmapsto x + \varepsilon h_1$$
$$y \longmapsto y + \varepsilon h_2$$
$$xy + \varepsilon g \longmapsto (x+\varepsilon h_1)(y+\varepsilon h_2) + \varepsilon g$$
$$= xy + \varepsilon(h_1 y + h_2 x + g)$$

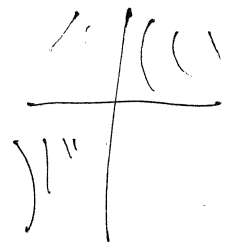
A: Yes, take $h_1 = 0$ $h_2 = 1$.

In this case we get an iso

$$\mathrm{Def}_A(R) \cong \cancel{A/(x,y)} \not\cong \mathbb{C} \left(= A/(x,y)\right)$$

$$R[x,y]/(xy+\varepsilon g) \longleftarrow g$$

This fits also our geometric intuition.



semifree: free as a graded alg,

not (nec) free as a dg alg.

Before checking the semifree, Let's check what does freeness mean!

Consider the forget functor

$$\mathcal{U}: \text{Comm-dg}\,\text{Alg}_k^{\leq 0} \longrightarrow \text{dg}\,\text{Vect}_k^{\leq 0} = \text{Ch}_k^{\leq 0}$$

↑ chain complex.

The free functor F will satisfy

$$\text{Hom}_{\text{CDGA}}(F(V), A) \cong \text{Hom}_{\text{Ch}_k}(V, \mathcal{U}(A))$$

E.g. When $V = kx \oplus ky$, $\deg x = \deg y = 0$, $dx = dy = 0$,

then $F(V) = k[x,y]$, $df \equiv 0$.

When $V = kx \oplus ky \oplus ke$, $\deg x = \deg y = 0$, $dx = dy = 0$

$\deg e = -1$ $de = x + y$

then $F(V) = k[x,y] \otimes \Lambda(e)$ $df \equiv 0$

$= k[x,y] \oplus k[x,y]e$ $de = x + y$

If we take $P = k[x,y] \otimes \Lambda(e)$ $\deg x = \deg y = 0$ $de = xy$

$\deg e = -1$

then P is not free, but only semifree.

Right side.

Still, set $A = k[x_1, \dots, x_n]/(f_1, \dots, f_r)$, $\operatorname{char} k = 0$.

Prop. $\exists$ semifree dg alg P where

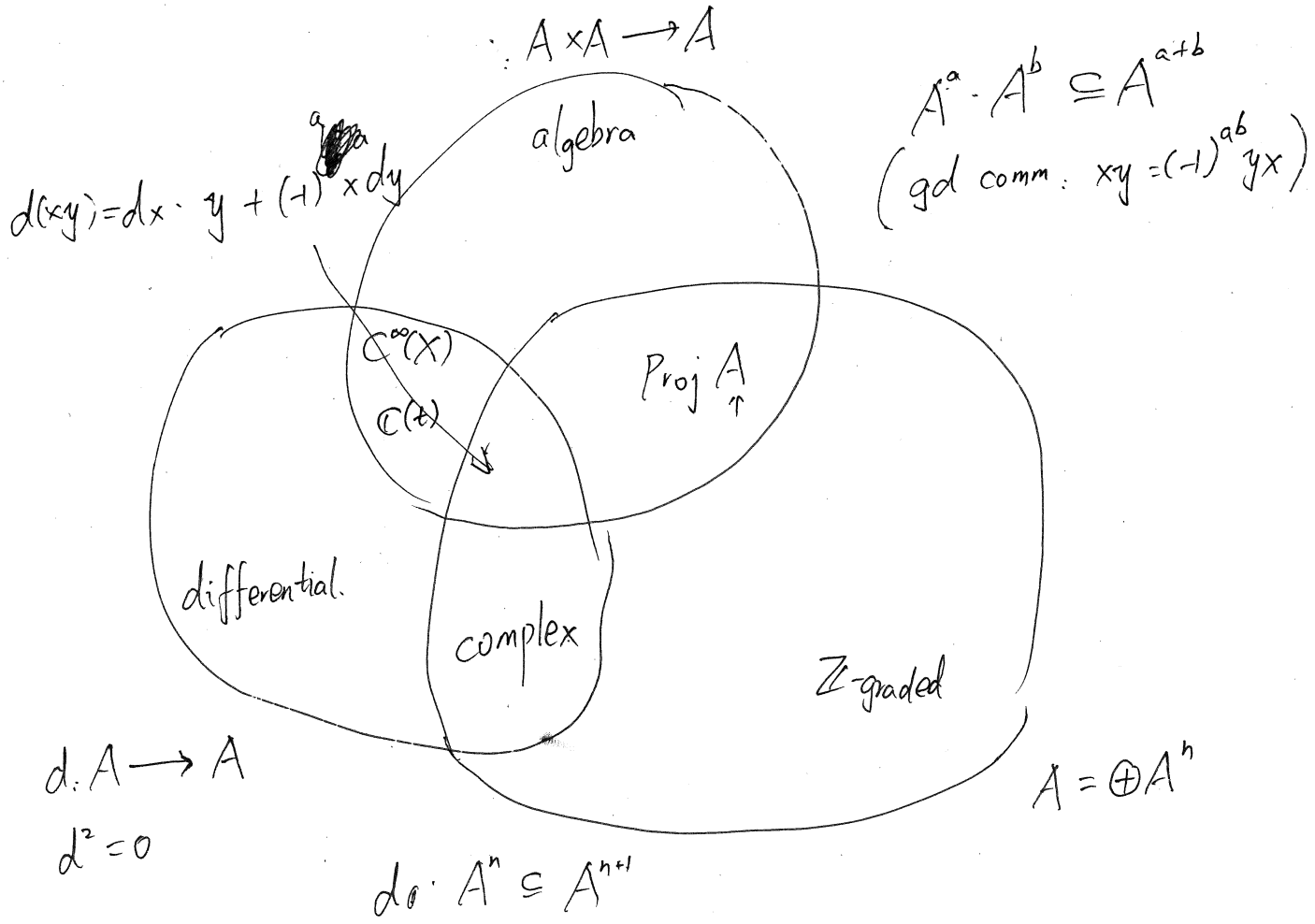
$$H^i(P) = \begin{cases} A & i = 0 \\ 0 & i \neq 0 \end{cases}$$

(i: $\mathbb{Z}$, k: k)

slogan: P is a resolution of A

Def. $\mathbb{L}_{A/k} = A \otimes_P \Omega_{P/k}$. is the cotangent complex

There are lots of new concepts to explain.



Def $P \in \text{Comm-dgAlg}_k^{\leq 0}$ is semifree, if

- P is a free graded alg generated by homo elements $x_1, x_2, \cdots$, and
- $dx_i \in \langle x_1, \cdots, x_{i-1} \rangle_{gr} Alg_k$

Eg When $P = k[x,y] \otimes \Lambda(e)$ $\deg x = \deg y = 0$ $df \equiv 0$

$\deg e = -1$ $de = xy$.

P is semifree, and

$$H^i(P) = \begin{cases} k[x,y]/(xy) & i=0 \\ 0 & i \neq 0 \end{cases}$$

Eg. When $A = k[x,y,z]/(xy, yz, zx)$, we construct P step by step:

$P_0 = k[x,y,z]$ $\deg f = 0$ $df = 0$.

$P_1 = P_0[e_1, e_2, e_3]$ $\deg e_i = -1$ $de_1 = xy$ $de_2 = yz$ $de_3 = zx$

$P_2 = P_1[u_1, u_2]$ $\deg u_i = -2$ $du_1 = xe_1 - ye_2$

$du_2 = ye_2 - ze_3$

$$\begin{pmatrix} -xy & yz \\ zx & -yz \end{pmatrix}$$

$$P_1 = \underset{-3}{P_0 e_1 e_2 e_3} \xrightarrow[d^{-3}]{\begin{pmatrix} yz \\ zx \\ xy \end{pmatrix}} \underset{-2}{P_0 e_2 e_3 \oplus P_0 e_3 e_1 \oplus P_0 e_1 e_2} \xrightarrow[d^{-2}]{\begin{pmatrix} 0 & yz & -yz \\ -zx & 0 & zx \\ xy & -xy & 0 \end{pmatrix}} \underset{-1}{P_0 e_1 \oplus P_0 e_2 \oplus P_0 e_3} \xrightarrow[d^{-1}]{(yz,\ zx,\ xy)} \underset{0}{P_0}$$

$$d(e_2 e_3) = de_2 \cdot e_3 - e_2 de_3$$
$$= \underset{zx}{e_3} - \underset{xy}{e_2}$$

$$d(e_1 e_2 e_3) = de_1 \cdot e_2 e_3 - e_1 \cdot d(e_2 e_3)$$
$$= de_1 \cdot e_2 e_3 - e_1 \cdot de_2 \cdot e_3 + e_1 e_2 de_3$$
$$= \underset{yz}{e_2 e_3} + \underset{zx}{e_3 e_1} + \overset{xy}{e_1 e_2}$$

For $(h_1, h_2, h_3) \in \ker d^{-1}$,

$$yz\,h_1 + zx\,h_2 + xy\,h_3 = 0$$

$$\Rightarrow \begin{cases} yz\,h_1(0, y, z) = 0 \\ \cdots \end{cases} \Rightarrow \begin{cases} x \mid h_1 \\ y \mid h_2 \\ z \mid h_3 \end{cases}$$

$$\Rightarrow xyz\left(\frac{h_1}{x} + \frac{h_2}{y} + \frac{h_3}{z}\right) = 0$$

generators are:

$$(x, -y, 0) \quad \& \quad (0, y, -z)$$
$$\| \qquad\qquad \|$$
$$x e_1 e_2 - y e_2 e_3 \qquad y e_2 e_3 - z e_3 e_1$$
$$\| \text{def} \qquad\qquad \| \text{def}$$
$$u_1 \qquad\qquad u_2$$

Relations are:

$$d(e_1e_2) = yz\,de_2 - zx\,de_1 = 0 - zu_1$$

$$d(e_2e_3) = zx\,de_3 - xy\,de_2 = 0 - xu_2$$

so $H^{-1}(P_1) = P_0u_1 \oplus P_0u_2 / \langle -zu_1, -xu_2 \rangle$ as P_0-module.

Rmk. For $S = k[x_1, \dots, x_n]$, $\varphi: S^r \longrightarrow S^m$ S-linear,

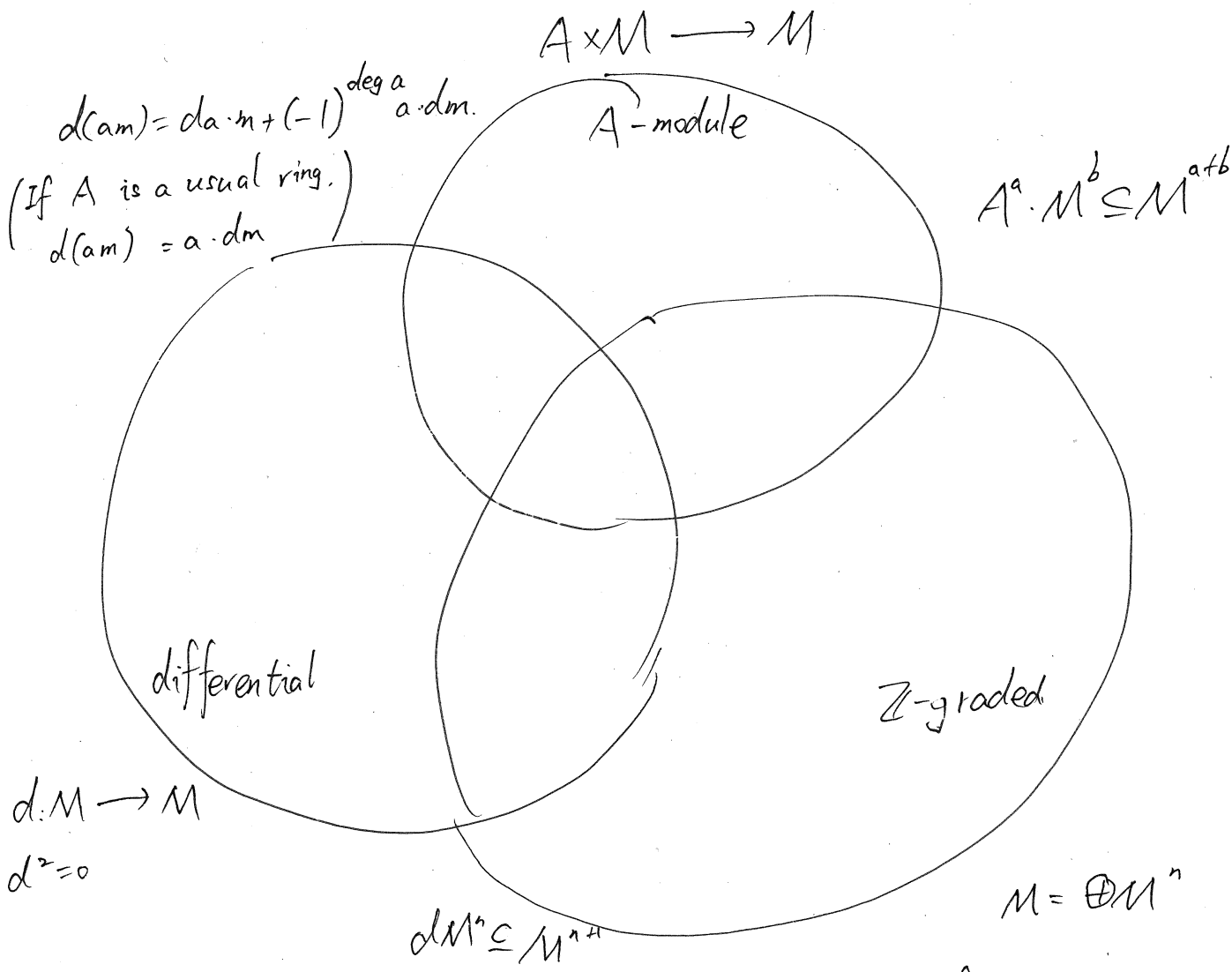
Hilbert's basis thm $\Rightarrow$ $\ker\varphi$ is f.g.

Schreyer's algorithm $\Rightarrow$ How $\ker\varphi$ is computed.

Rmk. $A = H^0(P)$ and $\Omega_{P/k}$ are all dg P-module, and

$\mathbb{L}_{P/k} = A \otimes_P \Omega_{P/k}$ is a dg A-module, i.e.,

a cplx of A-modules.



dg P-module structure on A: $\deg f = 0$ $df = 0$

$$P \longrightarrow A$$
$$x_i \longmapsto x_i$$
$$e_i \longmapsto 0$$

dg P-module structure on $\Omega_{P/k}$.

$$\deg \delta f = \deg f \qquad d\delta f = \delta df$$

E.g. For $A = k[x,y]/(xy)$, $\quad P = k[x,y] \otimes_k \Lambda(e) \quad de = xy$,

$$\Omega_{P/k} = P\delta x \oplus P\delta y \oplus P\delta e / (2e\delta e)$$

↳ we use δ to avoid the confusion with

where $\deg \delta x = \deg \delta y = 0$, $\deg \delta e = -1$,

$d\delta x = d\delta y = 0 \qquad d\delta e = \delta de$

$= \delta(xy)$

$= x\delta y + y\delta x$.

$$\mathcal{U}(\Omega_{P/k}) = 0 \longrightarrow Re\,\delta e \xrightarrow{d} \begin{matrix} Re\,\delta y \\ \oplus \\ R\,\delta e \end{matrix} \xrightarrow{\begin{pmatrix} xy & \\ & xy \\ y & x \end{pmatrix}} \begin{matrix} R\delta x \\ \oplus \\ R\,\delta y \end{matrix} \longrightarrow 0$$

$(-y \; -x \; xy)$

$$d(e\delta e) = de \cdot \delta e - e\,d\delta e$$

$$= xy\delta e - xe\delta y - ye\delta x$$

$$\mathbb{L}_{A/k} = A \otimes_P \Omega_{P/k} = A\delta x \oplus A\delta y \oplus A\delta e$$

$$\mathcal{U}(A \otimes_P \Omega'_{P/k}) = \left[0 \longrightarrow A\delta e \xrightarrow{d = \begin{pmatrix} y \\ x \end{pmatrix}} \begin{matrix} A\delta x \\ \oplus \\ A\delta y \end{matrix} \longrightarrow 0 \right]$$

$$\begin{matrix} -1 \\ \\ 0 \end{matrix}$$

$$f\delta e \longmapsto df \otimes \delta e + f\, d\delta e$$

$$= xf\delta y + yf\delta x$$

$$\Rightarrow H^i(\mathbb{L}_{A/k}) = \begin{cases} \Omega_{A/k} & i = 0 \\ k[x,y]/(x,y) & i = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$RHom(\mathbb{L}_{A/k}; A) = \left[0 \longleftarrow A\partial e \xleftarrow{(y\ x)} \begin{matrix} A\partial_x \\ \oplus \\ A\partial_y \end{matrix} \longleftarrow 0 \right]$$

$i \neq 0$.

$$\begin{matrix} -1 \\ \\ 0 \end{matrix}$$

$$Ext^i(\mathbb{L}_{A/k}; A) = \begin{matrix} A/(x,y) & A \cdot (y\partial_x - x\partial_y) \\ i = 1 & i = 0 \end{matrix}$$

Rmk. When $A = k[x_1, \dots, x_n]/(f_1, \dots, f_r) \cong S/I$.

$f_1, \dots, f_r$ is a regular sequence, then

$$P = k[x_1, \dots, x_n] \otimes \Lambda(e_1, \dots, e_r) \qquad \deg x_i = 0 \quad \deg e_i = -1$$
$$d x_i = 0 \quad d e_i = f_i$$

is a ~~free~~ semifree resolution, and

$$\begin{array}{ccccc} \oplus A f_i & & \oplus A e_i & & \\ \| & & \| & & \\ \mathbb{L}_{A/k} \cong [A^r & \xrightarrow{J = \left(\frac{\partial f_i}{\partial x_j}\right)_{i,j}} & A^n]^{\sim} & & \\ \| & & \| & & \\ I/I^2 & \xrightarrow{d} & i^* \Omega^1_{S/k} & & \end{array}$$

This P is called the Koszul ~~algebra~~ DG algebra.

After changing the degree notation, it is a Koszul c.g.a.
(bigraded algebra)

We always have a map

$$\mathbb{L}_{A/k} \longrightarrow \Omega_{A/k}$$

with iso on $H^0(-)$, so

$$\mathbb{L}_{A/k} \cong \Omega_{A/k} \iff H^i(\mathbb{L}_{A/k}) = 0 \quad \forall i < 0.$$

Furthermore, when $\operatorname{Spec} A$ is reduced & lci, $\mathbb{L}_{A/k} \cong \Omega_{A/k}$

$$\ker J = \left\{ \underset{(a_1, \dots, a_r)}{v} \in A^r \;\middle|\; \sum_j \frac{\partial f_i}{\partial x_j}(a_j) = 0 \right\}$$

$\forall \mathfrak{p} \subseteq A$ minimal, $a_i \in \mathfrak{p} \overset{\text{reduced}}{\Longrightarrow} a_i = 0$

lci $\Rightarrow$ rk ~~Supp ker~~ $J_\eta = r$. $\Rightarrow$ $N_\eta \in A_\eta^r$ is 0.

$\Rightarrow$ $a_i \in P$.



In fact, we obtain $\mathbb{L}_{A/k} \simeq [\, I/I^2 \xrightarrow{d} i^*\Omega_{S/k} \,]$

$\quad\quad\quad

Eg. [2406.02440, Lemma 3.1]

[2606.16829, Example 8.8]

Let $A_n = \dfrac{k[x_1, \dots, x_n]}{(x_i x_j : i \neq j)}$ $\quad T^i(A_n) = Ext^i_{A_n}(k_{A_n/k}, A_n)$

	A_2	A_3	A_4	
T^0	Ω_{A_2}	Ω_{A_3}		
T^1	0	$k^{\oplus 3}$	$k^{\oplus 8}$	$n(n-2)$
T^2	0	0	$k^{\oplus 6}$	$\binom{n}{2}(n-3)$ $n \geq 2$
T^3	0	k	$k^{\oplus 12}$	$\binom{n}{3}(2n-5)$ $n \geq 3$
T^4	0	$k^{\oplus 3}$	$\dim \geq 24$	

$\sum \dim_k T^i(A_3)\, t^i = \frac{2-t}{1+t} \sum_{n \geq 1} c_n t^n - t$

$= 3t + t^3 + 3t^4 + 6t^5 + 10t^6 + 15t^7 + \cdots$

$c_n = \frac{1}{n} \sum_{d \mid n} (-1)^{n+n/d} \mu(d)\, 2^{n/d}$

I don't think it vanishes when $i \gg 0$.

(~~A_3~~ A_3 case proved by math/9809132. (5.2).)

Conj. (Herzog's last theorem)

Let $\operatorname{char} k = 0$, $\quad A = k[x_1, \dots, x_n]/I$.

Write $T^i(A) = Ext^i_A(k_{A/k}; A)$, then.

$T^i(A) = 0$ for all $i \gg 0 \Rightarrow T^i(A) = 0$ for all $i \geq 2$

$$E_2^{p,q} = H^p(X,\ \mathcal{E}xt^q_X(\mathcal{F},\mathcal{G})) \Rightarrow \mathcal{E}xt_X^{p+q}(\mathcal{F},\mathcal{G})$$

canonical truncation filtration (Posthikov filtration)

Rmk. $\mathcal{E}xt^i(\mathcal{F},\mathcal{G}) \neq \Gamma(\mathcal{E}xt^i(\mathcal{F},\mathcal{G}))$

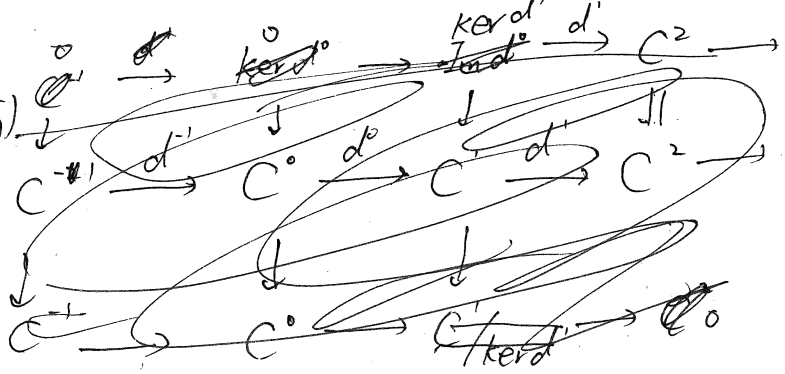
$\mathcal{E}xt^i(\mathcal{F},\mathcal{G})$ = $H^i(R\pi_* \circ R\mathcal{H}om(\mathcal{F},\mathcal{G}))$

$\Gamma(\mathcal{E}xt^i(\mathcal{F},\mathcal{G}))$ = $R^0\pi_* R^i\mathcal{H}om(\mathcal{F},\mathcal{G})$ = $H^0(\pi_* R^i\mathcal{H}om(\mathcal{F},\mathcal{G}))$

Suppose $R^i\mathcal{H}om(\mathcal{F},\mathcal{G}) = 0$ when $i \neq 0, 1$, then we get a SES

$$0 \to R^0\mathcal{H}om(\mathcal{F},\mathcal{G}) \to R\mathcal{H}om(\mathcal{F},\mathcal{G}) \to R^1\mathcal{H}om(\mathcal{F},\mathcal{G})[1] \to 0$$

$\mathcal{E}xt^0(\mathcal{F},\mathcal{G})$ = $\mathcal{H}om(\mathcal{F},\mathcal{G})$ = $\ker d^0$ = $\ker d'$ = $\mathcal{E}xt^1(\mathcal{F},\mathcal{G})[1]$



$$C^{-2} \to C^{-1} \to \ker$$

$$C^{-2} \to C^{-1} \to C^0 \to C^1 \to$$

$$\text{coker} \to C^1$$

which induces a LES.

$$\to H^2(\mathcal{H}om(\mathcal{F},\mathcal{G})) \longrightarrow \cdots$$

$$\to H^1(\mathcal{H}om(\mathcal{F},\mathcal{G})) \longrightarrow H^1(R\mathcal{H}om(\mathcal{F},\mathcal{G})) \longrightarrow H^0(\mathcal{E}xt^1(\mathcal{F},\mathcal{G}))$$

$$0 \longrightarrow H^0(\mathcal{H}om(\mathcal{F},\mathcal{G})) \longrightarrow H^0(\mathcal{H}om(\mathcal{F},\mathcal{G})) \longrightarrow H^1(\mathcal{E}xt^1(\mathcal{F},\mathcal{G})) \to 0$$

When $\mathcal{F} = \mathbb{L}_{X/k} \overset{\text{in our case}}{=\!=} \Omega_{X/k}$, $\mathcal{G} = \mathcal{O}_X$,

$$\mathcal{T}_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X).$$

$$\mathcal{T}_X^1 = \mathcal{E}xt^1(\Omega_{X/k}, \mathcal{O}_X).$$

$$\to H^2(\mathcal{T}_X) \longrightarrow \cdots$$

$$0 \to H^1(\mathcal{T}_X) \longrightarrow \mathcal{E}xt^1(\mathbb{L}_{X/k}, \mathcal{O}_X) \longrightarrow H^0(\mathcal{T}_X^1)$$